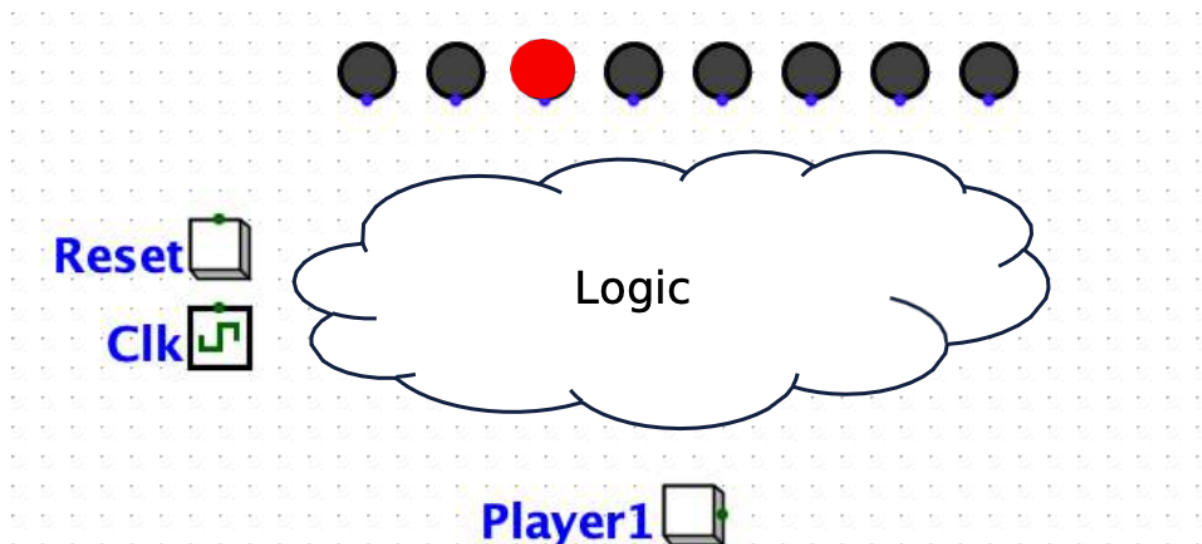


Implementation of Display Board, Lab Report (2nd part)

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Digital Systems Laboratory



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Introduction

Bigger Picture

The **display subsystem** is the visual part of the Pong project and is split in two lab sessions: session 2 (decoder+leds) and session 3(counter+clock+outside interfaces). Subsystem's role is to represent the current position of the ball on a line of eight LEDs, so that each clock cycle corresponds to a visible movement by one position. In hardware terms, the display is completed by combining the programmable **clock**, the 74191 up/down **counter**, the 74155 **decoder**, the **LEDs**, and the required resistors and power connections. Thus, the display acts as the interface between the future control logic and the user, transforming internal **binary position** data into a directly **observable LED pattern**. Advancing from what we completed in the previous Lab #2, in this Lab, the purpose is to complete this display board by **adding the counter and the corresponding external control inputs** to the decoder-LED subsystem already built in the previous laboratory stage.

A. Interface description as implemented

The connections/interface implemented on the full display circuit can be categorized as: 1) internal interconnections (ttl's connections to each other and to other circuit's elements) and 2) external interfaces (circuit's elements to the outside world or the opposite). The 1st ones mainly include:

- Counter - decoder connections (safely passing the 3 bits-digits from cnt to dec)
- Clock – counter connection
- Decoder – leds connections

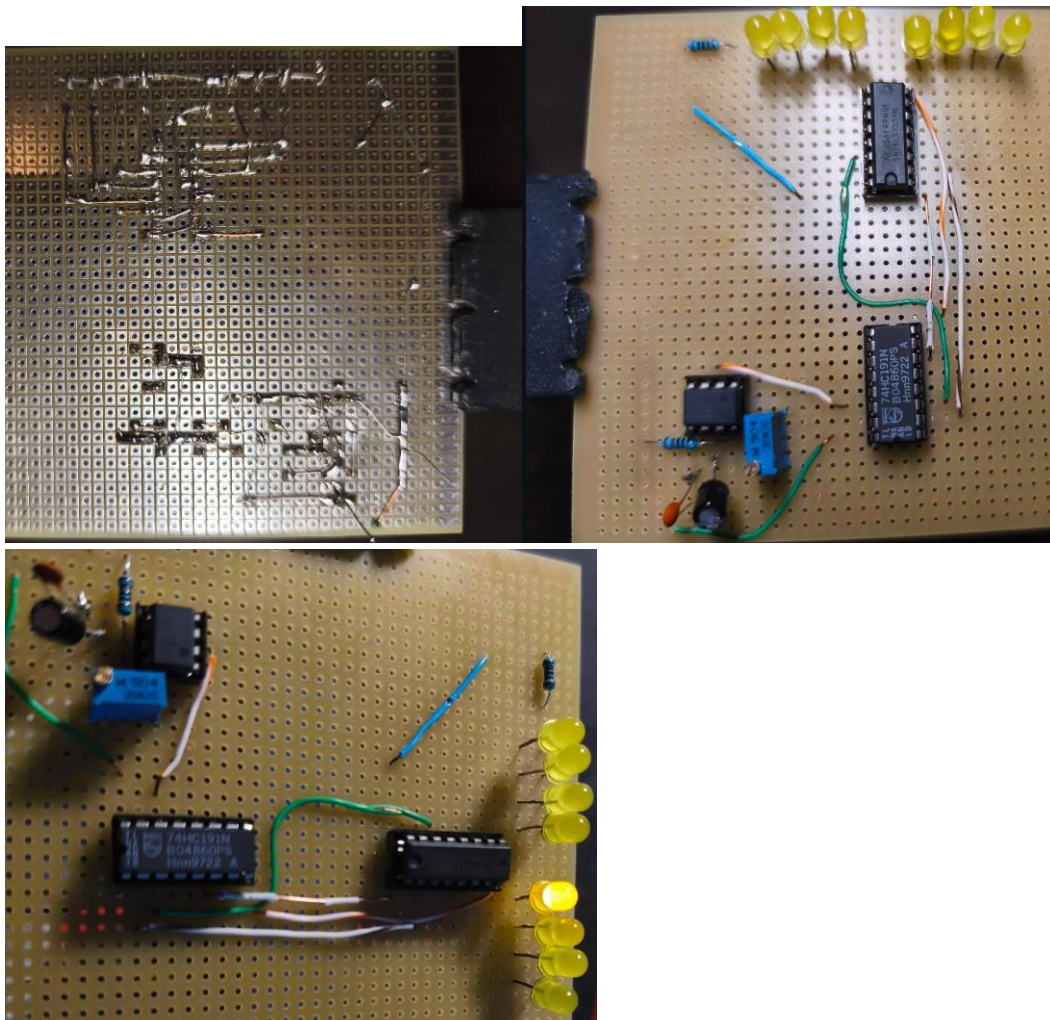
Whereas the 2nd ones are about outside signals, which are described in the array that follows:

<u>SigName</u>	<u>Direction</u>	<u>Width</u>	<u>Role</u>	<u>Hardw. Map</u>	<u>Polarity comment</u>
Vdd	Input	1bit	5V supply	Rail feeding clock, counter, decoder, leds	-
Gnd	Input	1 bit	Ground reference	Rail feeding clock, counter, decoder	-
Reset	Output	1 bit	Start again button -> 000	Button -> pin 11 of cnt	Logic 0 when pressed
Play	Output	1 bit	"Racket" Button	Not configured yet	Not configured yet
Clock_in	Input	1 bit	Clock pulse	Goes to 74191, pin 14	-
Core_reset	Input	1 bit	Reset to 000	Goes to 74191 pin 11	Active low
UpDown	Input	1 bit	Counting direction control	Goes to 74191 pin 5	LOW = count up HIGH = count down
EnDec/EnCnt	Input	1 bit each	Freezing dec/cnt state	Goes to 74155 pins 2, 14 / 74191 pin 4	Active low
EdgeL	Output	1 bit	Sign ball reached Left Edge	Not implemented yet	Active High
EdgeR	Output	1 bit	Sign ball reached Right Edge	Not implemented yet	Active High
Clock_out	Output	1 bit	Clock pulse	From 555 pin 3	-

This reports implementation is complete regarding internal interconnections and interfaces, making sure the main building blocks – TTLs of the subsystem “Display” are effectively connected to each other and to the main outside interfaces (Vdd, GND, Clock_in, Core_reset, UpDown, Clock out). However, the interfaces Reset, Play, Edge L and Edge R require future laboratory sessions in order to be effectively soldered and shown in picture. Theoretically, though, we plan on implementing the following:

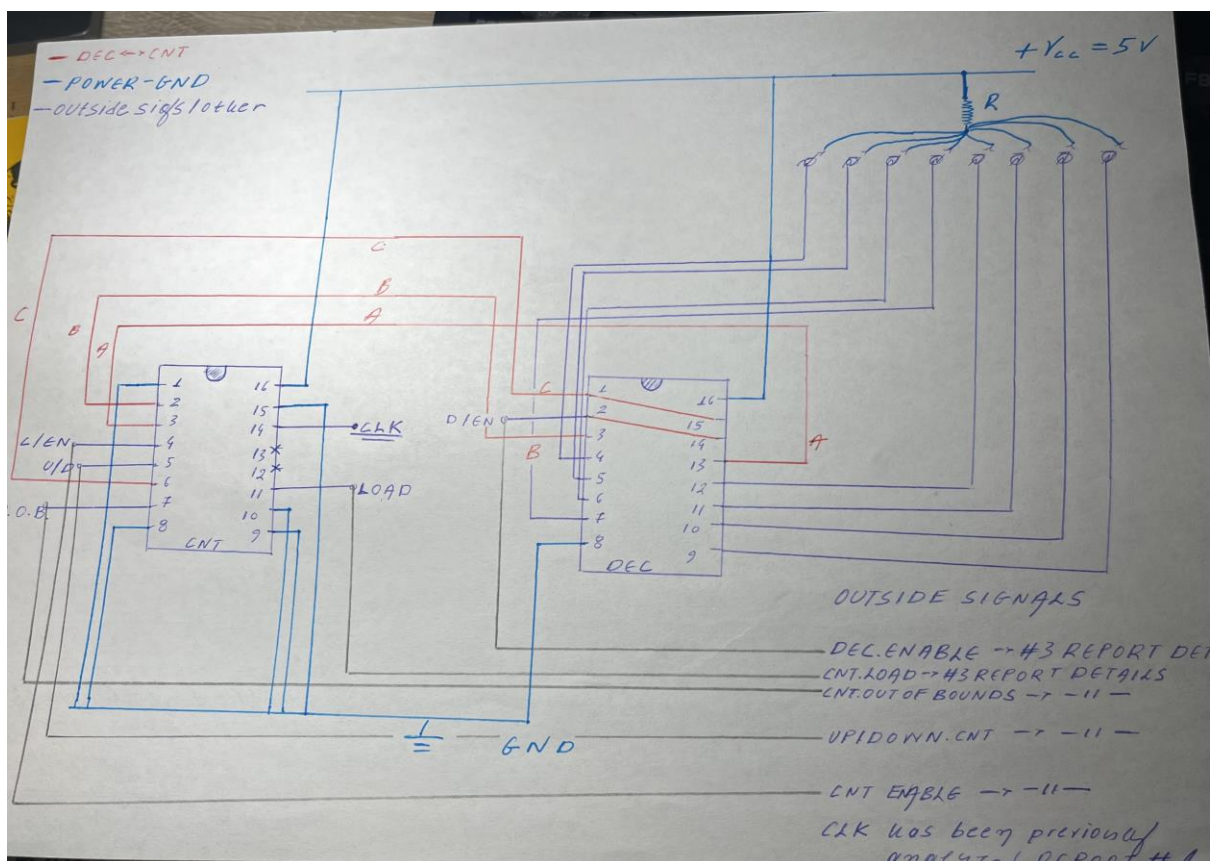
- **RESET SWITCH:** Button connected to LOAD – pin 11 of counter TTL. LOAD has active low logic, so it needs 0 from the button to force reset. Button gives 0 when manually pressed, otherwise gives 1.
- **PLAY :** Needs F-F for implementation, future labs will configure it.
- **Edge L/ Edge R:** For active high implementation needs inverters. Briefly, take Led0, Led7 signals and invert them, then display EDGE-L=1 | EDGE-R=1.

The core of the display: clock -> counter -> decoder -> LEDs can be seen below soldered on a perfboard:



B. Interface analysis and handwritten sketch

The circuit's subsystem "Display" requires cautious and specific interconnections between the elements used, since they are the foundation of the pong's operation. The hand-written sketch pasted below is the base that led to our breadboard and finally soldered implementation. It clearly shows the interconnections chosen by us to create the needed result:



Interface, interconnections breakdown:

- Decoder's** properties and connections to the Leds were analyzed in the previous Lab.2 report. It is important to emphasize its active low logic, causing LEDs to be connected to the Vcc as well. Furthermore, we should state again its operation as a 3 to 8 decoder, thanks to pins 1,15 (carry bit C) | pins 2, 14 (common enable) being tied together. Dec's enable is also active low, meaning reception of 0 -> normal operation and 1 -> OFF.
- Counter's** connections can be analyzed as follows:
Clock's pulse comes in pin 14 and causes changes of states (counting).
Binary number DCBA is formed in pins 7, 6, 3, 2. MSB D is kept aside, whereas the other 3 are sent to the dec.
Cnt's Enable is active low, meaning: 0->counting, whereas 1->frozen.
Out of bounds MSB is probably going to be used later in control block.
LOAD is resetting to 0000 (grounded data A, B, C, D) when given 0.
Up/Down logic is : 0->count up and 1->count down.

C. Expected, tested behavior

At the present stage of the project, the **full Display subsystem** has been successfully implemented on perfboard, consisting of the programmable clock, the 74191 up/down counter, the 74155 decoder, the LED display array, the current-limiting resistor network, and the required power and ground connections. The role of this subsystem is to represent the current ball position as a 3-bit count value and to decode this value into the activation of exactly one LED at a time. In Lab 3, the goal is precisely to complete this hardware display board by adding the counter and its control inputs to the previously implemented decoder–LED stage, while the actual Control logic is still external and will be implemented later.

In the present prototype, the board was tested in a manual but fully meaningful way. Before operation, the circuit is powered and the required control signals are externally forced to valid logic levels. In normal display operation, LOAD/Core_reset is kept inactive (1), En is kept active (0), and the counting direction is selected by the user through the UpDown signal. Under these conditions, the 74191 updates its 3 least significant bits at each clock pulse, and the 74155 decodes the resulting value so that exactly one LED is illuminated at a time. In this way, the visible LED position advances or retreats deterministically through the 8 allowed display states. The result is a correct and stable realization of the Full-Display subsystem of the Pong project, even though the control signals are not yet generated automatically by dedicated game logic.

When Core_reset is asserted, the counter is forced back to its predefined initial state and the display returns to the initial LED position. Likewise, reversing the UpDown signal reverses the sequence of illuminated positions, while deactivating En suppresses the visible display. Therefore, the current board already performs the essential display task correctly: it can initialize, count in both directions, and show one unique ball position at every clock cycle. The prototype is therefore functionally complete **as a display subsystem**, even though it is not yet a fully autonomous Pong system.

D. Experience, implementation notes

The implementation process clearly showed the importance of dividing the design into smaller, verifiable stages. The transition from breadboard experimentation to the final perfboard realization confirmed that correct hardware behavior depends not only on the logic design itself, but also on careful pin identification, clean wiring, and systematic verification of each individual subsystem before final soldering. In particular, the staged testing of the clock, counter, and decoder proved essential for avoiding cumulative errors during assembly. This is fully consistent with the Lab 3 instructions, which emphasize pin-level study, repeated checking before soldering, and experimentation with alternative arrangements before committing to the final board.

A particularly useful practical lesson was that the current implementation, although “manual” in its control, is already sufficient to validate the correctness of the display datapath. By assigning the external control signals deliberately and observing the resulting LED sequence, it became possible to confirm the

real hardware behavior of the board without yet needing the full game controller. This helped isolate the actual task of Lab 3: the completion and verification of the display board itself. Finally, the implementation made clear that good hardware design is not only about functional correctness, but also about preparation for later integration. The present board was built not as an isolated circuit, but as a subsystem whose interfaces are meant to connect to the future Control logic. In that sense, this stage was both a successful standalone implementation and an important foundation for the complete Pong system.

